

Name:		Index Number:		Class:	
-------	--	---------------	--	--------	--



DUNMAN HIGH SCHOOL
Preliminary Examination
Year 6

H2 BIOLOGY

9744/01

Paper 1 Multiple Choice Questions

25 September 2018

1 hour

Additional Material: OTAS sheet

INSTRUCTIONS TO CANDIDATES:

DO NOT TURN THIS PAGE OVER UNTIL YOU ARE TOLD TO DO SO.

READ THESE NOTES CAREFULLY.

There are **thirty** questions in this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

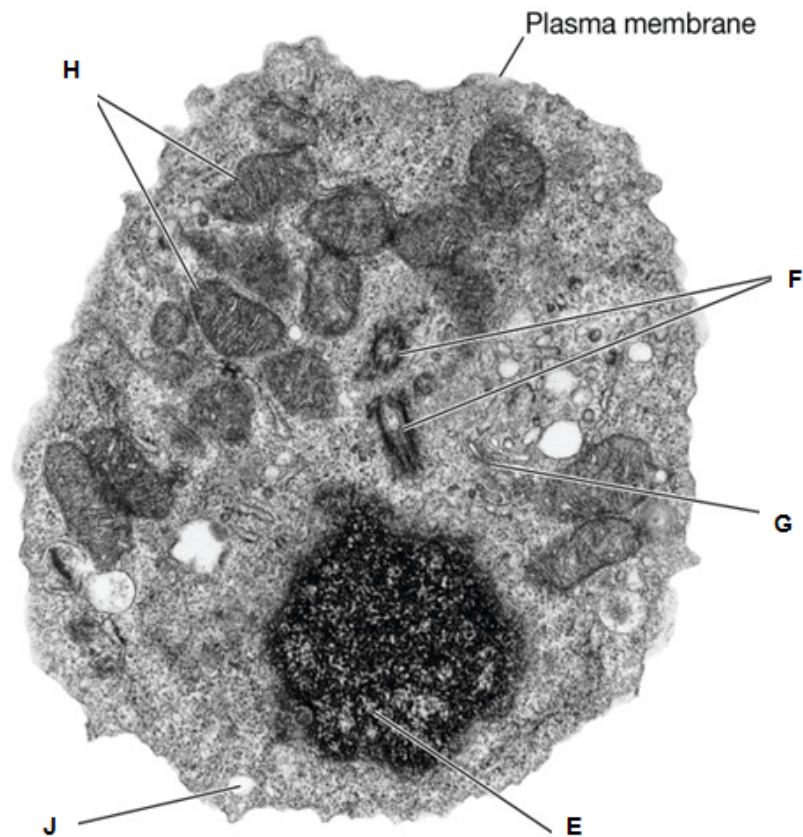
This document consists of **24** printed pages and **0** blank page.

[Turn over

Multiple Choice Questions (30 marks)

Answer **all** questions in this section.

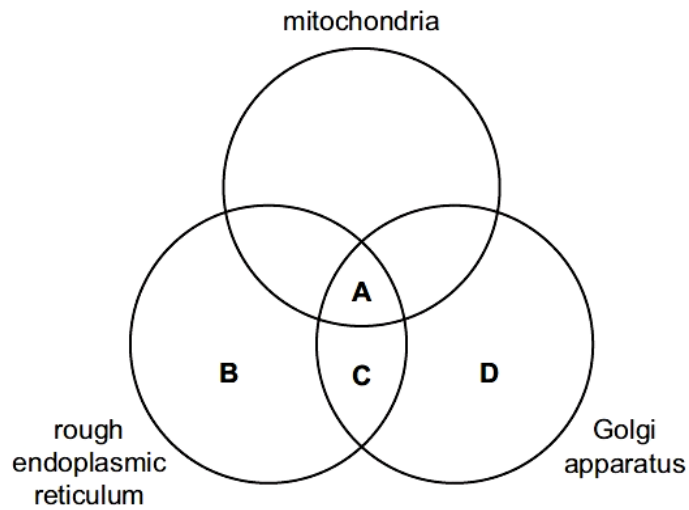
- 1 An electron micrograph of a cell is shown below.



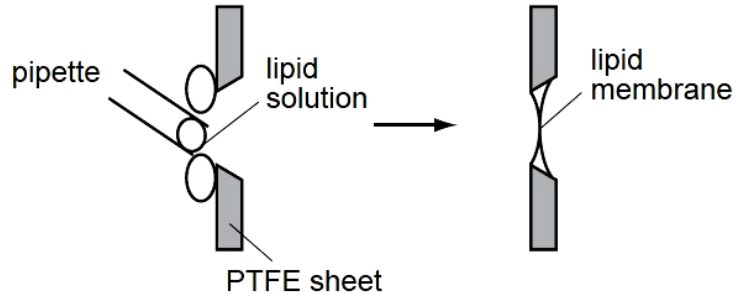
Match the organelles **E, F, G, H** and **J** associated with the cellular processes listed.

	E	F	G	H	J
A	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation	Packaging of secretory products
B	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication	Packaging of secretory products
C	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products	DNA replication
D	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation	Digestion of material

- 2 Which organelle(s) is / are required for the formation of the hydrolytic enzymes found in lysosomes?



- 3 Lipid membranes can be formed in the laboratory by painting phospholipids over a support (PTFE sheet) with a hole in it.



Such a lipid membrane is impermeable to water-soluble materials including charged ions such as Na^+ or K^+ . In one experiment with Na^+ ions, no current flowed across the membrane until a substance called gramicidin was added.

Which statement is consistent with this information and your knowledge of membrane structure?

Gramicidin becomes incorporated into the membrane and is _____.

- A a carbohydrate molecule found only on the outside of the membrane.
- B a non-polar lipid which passes all the way through the membrane.
- C a protein molecule with both hydrophilic and hydrophobic regions.
- D a protein molecule which has only hydrophobic regions.

4 Which of the following is / are **TRUE** of triglycerides and phospholipids?

- 1 Both contain glycerol.
- 2 Both are not amphipathic.
- 3 Triglycerides have a higher carbon content than phospholipids for storage purposes.
- 4 Phospholipids have hydrophilic regions to interact with cell cytoplasm unlike triglycerides.
- 5 Triglycerides are formed from glycerol and fatty acids but phospholipids are formed from glycerol and phosphoric acid only.

A 1, 3 and 5

B 1, 2 and 3

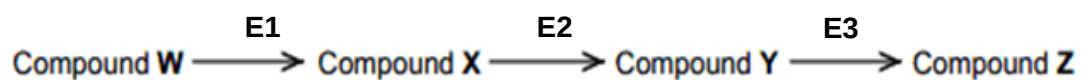
C 1, 3 and 4

D 2, 3 and 4

5 Which set of statements correctly describes haemoglobin?

A	Four polypeptide chains, each containing a prosthetic group	Iron ions can associate with oxygen forming oxyhaemoglobin	In each chain, hydrophobic R groups of amino acids point towards the centre of the molecule	At 50% saturation, two oxygen molecules are transported by the molecule
B	Polypeptide chains interact to produce a globular chain	Each chain contains a prosthetic group of amino acids surrounding an iron ion	Two identical alpha chains and two identical beta chains	Each chain can transport an oxygen molecule
C	Polypeptide chains interact to produce an almost spherical molecule	An iron ion is present within each haem group	Quaternary structure of two alpha chains and two beta chains	Each molecule can transport a total of four oxygen atoms
D	Polypeptide chains produce a loose helical shape, which curls to form a spherical molecule	Iron ions in the molecule can bind reversibly with oxygen	In each chain, hydrophobic R groups of amino acids surround the iron ion	Each molecule can transport a total of eight oxygen atoms

- 6 The flowchart below shows part of an enzyme-catalysed sequence which involves 4 compounds (W – Z) and 3 enzymes (E1 – E3).



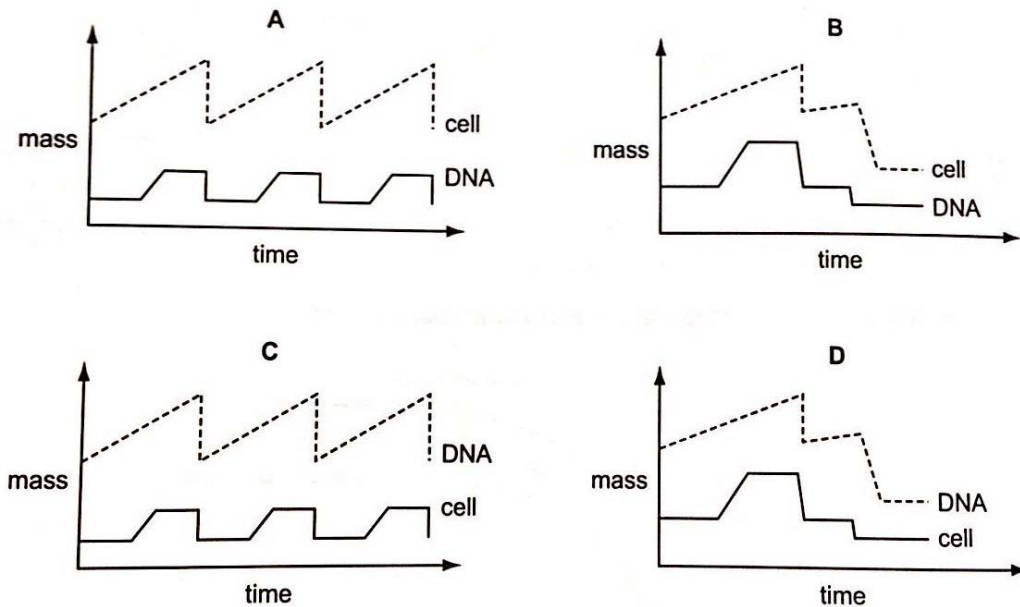
The addition of another compound M results in no change in the concentration of compound W, an accumulation of compound X, and a near absence of both compound Y and compound Z.

What does this information indicate about the mode of action of compound M, and which is a method by which the production of compound Z can be increased in the presence of compound M?

	Mode of action of compound M	Method to increase production of compound Z
A	It inhibits E1.	Increase concentration of compound X in the reaction mixture.
B	It activates E1.	Increase concentration of E2 in the reaction mixture.
C	It inhibits E2.	Increase concentration of compound Y in the reaction mixture.
D	It inhibits E3.	Increase concentration of compound Y in the reaction mixture.

- 7 Cell division is the means of almost all growth and reproduction.

Which graph correctly represents a form of cell division that maintains genetic stability at expense of variation?

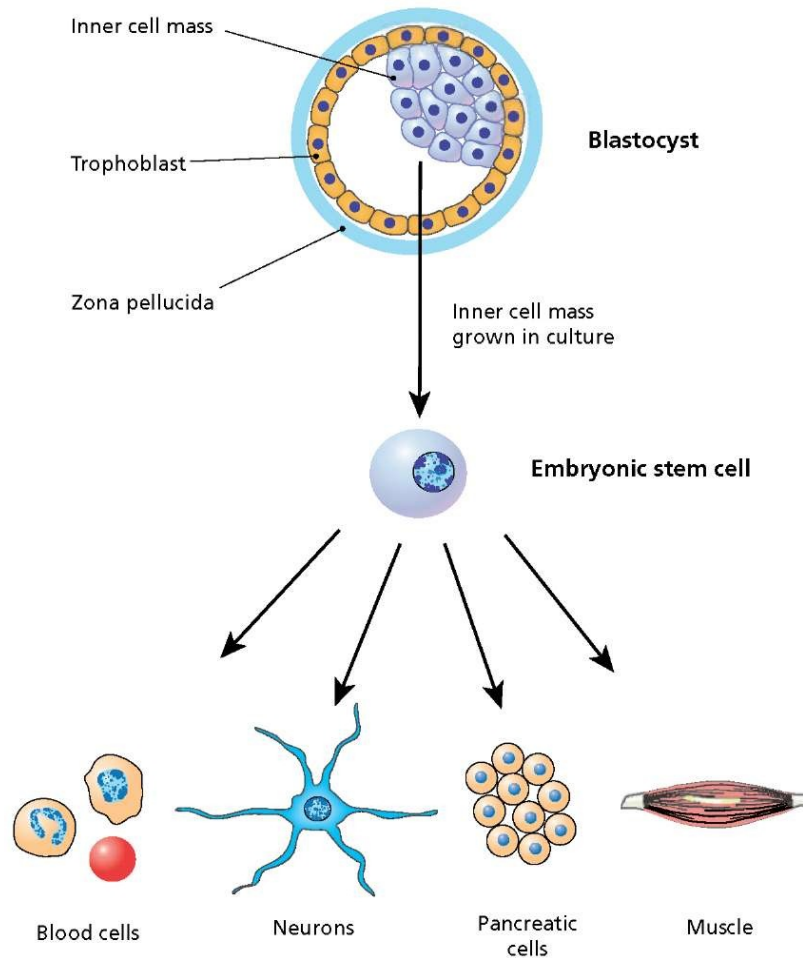


- 8 Which of the following statements about cancer are **TRUE**?

1. Oncogenes and tumor suppressor genes can both be detected by introducing fragmented DNA from cancer cells into suitable cell lines and isolating colonies that display cancerous properties
2. Individuals who inherit one inactive copy of tumour suppressor gene are more likely to develop cancer than individuals with two non-mutant copies
3. Viruses and other infectious agents play no role in human cancers
4. In the cellular regulatory pathways that control cell growth and proliferation, the products of oncogenes are inhibitory components and the products of tumour suppressor genes are stimulatory components
5. When analysed, cancer cells are often found to have only one mutation in a regulatory pathway that controls cell proliferation

- A 1 and 2
B 2 and 3
C 3 and 5
D 1, 2 and 4

- 9 Which of the following features of the embryonic stem cells and specialized cells shown in the diagram are **TRUE**?



	Embryonic stem cells	Specialized cells
A	Embryonic stem cells display greater plasticity when grown in culture than when in blastocyst.	The blood cells are genetically different from the embryonic stem cells but have shorter telomeres.
B	Embryonic stem cells are totipotent and are capable of differentiating into many different cell types.	The pancreatic cells are genetically identical to the embryonic stem cells but with a different set of genes expressed.
C	Embryonic stem cells are multipotent and are capable of differentiating into limited range of cell types.	The blood cells are genetically different from the embryonic stem cells because different genes are expressed.
D	Embryonic stem cells are pluripotent and are capable of differentiating into many different cells types.	The pancreatic cells are genetically identical to the embryonic stem cells but have shorter telomeres.

10 Two genes involved in coat colour of goats are at loci on different chromosomes.

The colour gene C causes the hairs to have uniform colour and has three alleles.

- C^{DB} giving dark brown hairs
- C^B giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^G) causes the development of white hairs between the coloured hairs giving the coat a shaded appearance.

The table shows the results of crosses between a male goat and two female goats.

parents	offspring
black agouti male x uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
black agouti male x medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	black agouti male	uniformly dark brown female	medium brown agouti female
A	$C^B C^{MB} A^G A^G$	$C^B C^{DB} A^G A^g$	$C^{MB} C^{MB} A^G A^G$
B	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^B C^{MB} A^G A^G$
C	$C^B C^{DB} A^G A^G$	$C^{DB} C^{MB} A^G A^g$	$C^B C^{DB} A^G A^g$
D	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^{MB} C^{MB} A^G A^G$

- 11** In rats, the allele of a gene for 'mottled' coat (M) and the recessive allele (m) for 'normal' coat are sex linked. The allele of a gene for 'long' whiskers (W) and the recessive allele (w) for 'short' whiskers are autosomal.

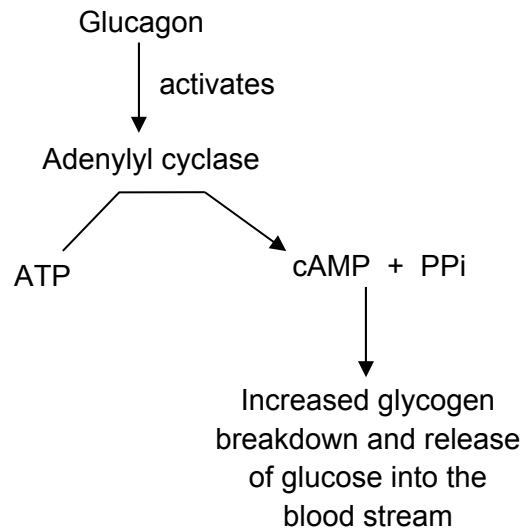
A male rat with a normal coat and short whiskers was mated on several occasions to the same female. The offspring showed the following phenotypes in equal proportions.

- Mottled females and males with long whiskers
- Mottle females and males with short whiskers
- Normal females and males with long whiskers
- Normal females and males with short whiskers

What are the genotypes of the parents?

- A** X^MYww and X^MX^MWW
- B** X^mYww and X^MX^MWw
- C** X^mYww and X^MX^mWW
- D** X^mYww and X^MX^mWw

- 12** One action of the hormone glucagon is to cause the activation of adenylyl cyclase in liver cells. Adenylyl cyclase is an enzyme that converts ATP into the messenger molecule cyclic AMP (cAMP), the presence of which triggers a cascade of events within the cell that leads to the breakdown of glycogen and the subsequent release of glucose into the blood stream. This pathway is summarised below.



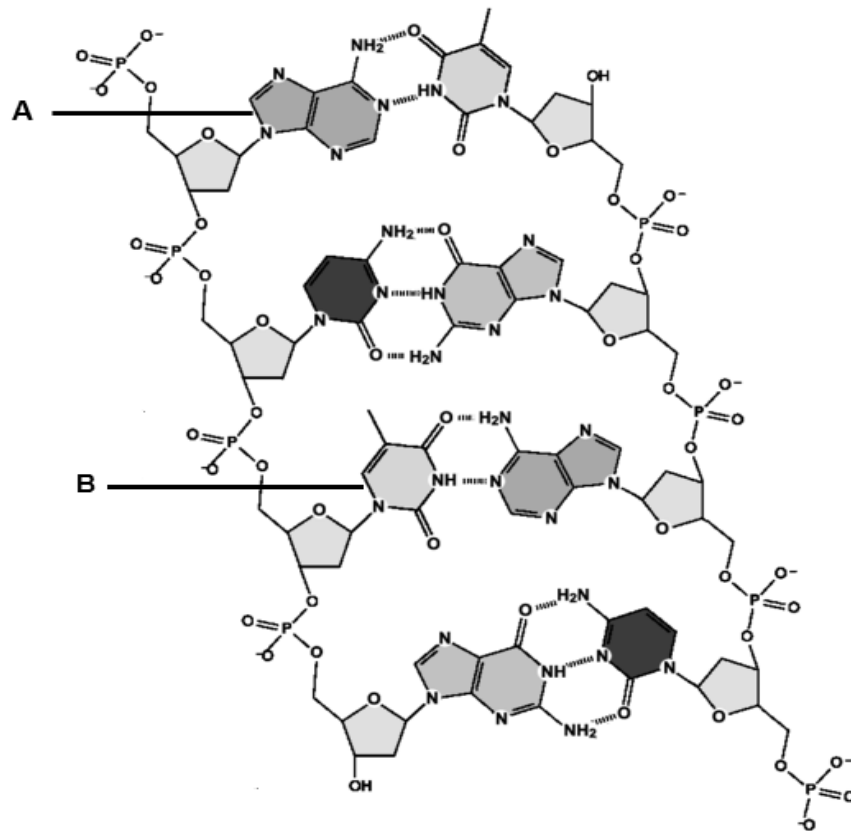
When glucagon is removed, adenylyl cyclase becomes inactive again and ceases to convert ATP to cAMP. The enzyme phosphodiesterase converts cAMP into AMP. The following table summarises an experiment measuring glucose release by liver cell cultures under various conditions.

Culture	Condition(s)	Observation(s)
1	no addition of glucagon and caffeine	minimal glucose release
2	glucagon added and later removed	glucose released into media; rapid reduction in release after glucagon removed
3	caffeine added	minimal glucose release
4	glucagon and caffeine added; glucagon was then removed but caffeine maintained	glucose released into media; slow reduction in release after glucagon removed

Based on these observations, which of the following statement is most likely to be true?

- A** Caffeine stimulates synthesis of cAMP.
- B** Caffeine prevents glucagon breakdown.
- C** Caffeine inhibits the action of phosphodiesterase.
- D** Caffeine binds the active site of adenylyl cyclase.

13 The figure below shows a DNA molecule.



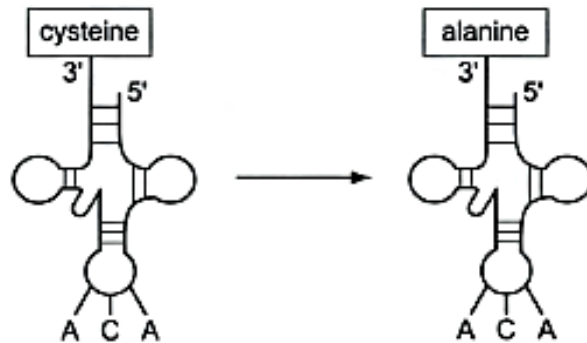
Which statement(s) correctly describe the polynucleotide?

1	The structure labelled A corresponds to that of a purine, while the structure labelled B corresponds to that of a pyrimidine.
2	The antiparallel nature of DNA double helix allows phosphodiester bonds to form between the nitrogenous bases of opposite strands.
3	Width of DNA double helix is $2\mu\text{m}$.
4	The wound DNA double helix consists of alternating major grooves and minor grooves along its axis.

- A 1 only
 B 1 and 4 only
 C 2 and 3 only
 D 1, 3 and 4 only

- 14** Transfer RNA combined with an amino acid is called amino-acyl tRNA. It is possible to chemically convert the amino acid cysteine into the amino acid alanine whilst it is still attached to its tRNA.

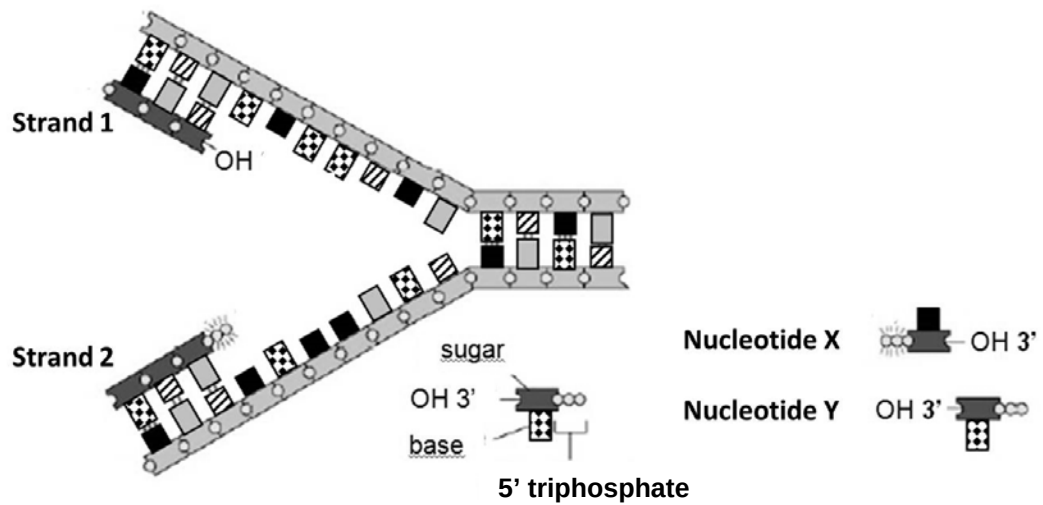
The altered amino-acyl tRNA still binds to UGU triplets on messenger RNA (mRNA), but now incorporates alanine into the resulting polypeptide instead of cysteine.



Which statement is correct?

- A** A codon on the amino-acyl tRNA determines its specificity.
- B** Both the amino acid and the anticodon of an amino-acyl tRNA affect where it binds to mRNA.
- C** The amino acid of an amino-acyl tRNA does not influence its binding to mRNA.
- D** The codon-anticodon interaction is influenced by the amino acid on an amino-acyl tRNA.

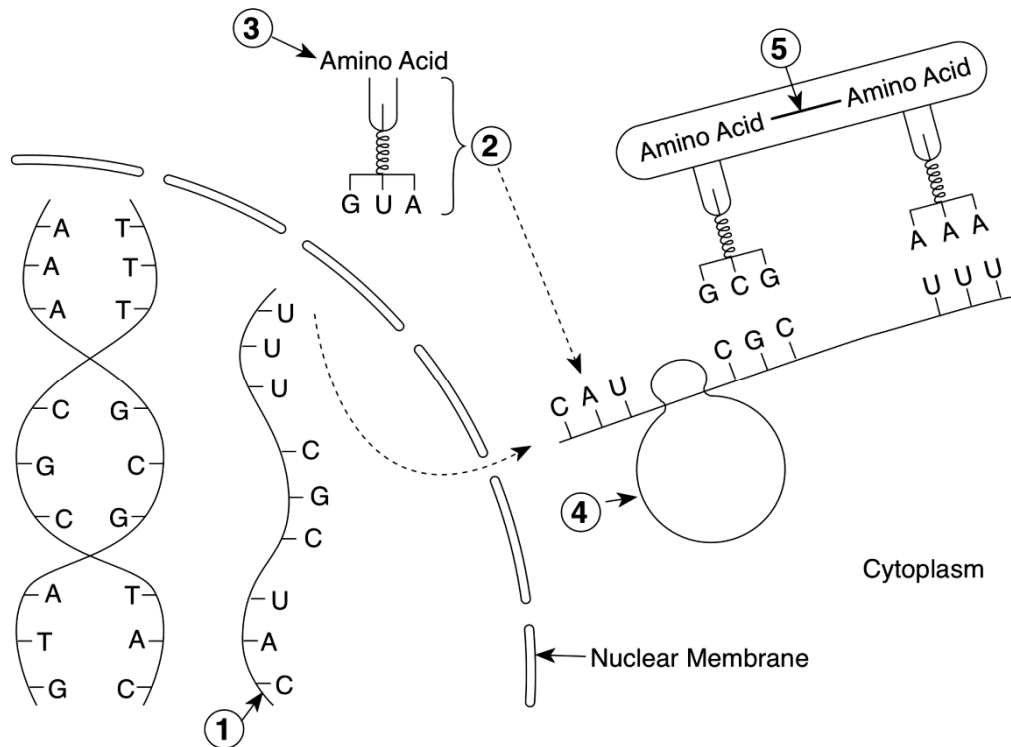
15 DNA replication is illustrated in the following figure.



Which of the following correctly describes the addition of the next nucleotide(s) in the DNA strands undergoing replication?

- A Nucleotide X will be added to the leading strand, which is strand 1.
- B Nucleotide Y will be added to the leading strand, which is strand 1.
- C Nucleotide X will be added to the lagging strand, which is strand 1.
- D Nucleotide Y will be added to the leading strand, which is strand 2.

16 The diagram represents some biochemical reactions involved in protein synthesis.



Which of the following is correct?

	The molecule coded directly from DNA is represented by	The bond labelled 5 is known as	The enzyme involved in catalysing bond 5 is
A	1 and 2	peptide	peptidyl transferase
B	1 and 2	peptide	aminoacyl tRNA synthetase
C	1, 2 and 3	peptide	peptidyl transferase
D	1, 2 and 4	carboxyl	aminoacyl tRNA synthetase

17 The following statements illustrate the processes that occur during translation.

- 1 The large subunit of the ribosome binds and forms the translation initiation complex.
- 2 The second amino acyl-tRNA complex now binds to mRNA at the “A” site of the ribosome.
- 3 The small ribosomal subunit, with initiator tRNA bound, binds to the 5’ cap of the mRNA and scans for the first start codon.
- 4 Soluble protein called release factor recognises the stop codon and binds at the “A” site.
- 5 Formation of a peptide bond between the first and the second amino acids by peptidyl transferase.
- 6 The second amino acyl-tRNA complex moves from the “A” site to the “P” site.

Which of the following shows the correct order?

- A** 3 → 1 → 2 → 5 → 6 → 4
- B** 1 → 5 → 2 → 3 → 6 → 4
- C** 3 → 5 → 2 → 1 → 4 → 6
- D** 1 → 5 → 2 → 3 → 4 → 6

18 Which of the following correctly matches the step involved in Southern blotting to its purpose?

	Step	Purpose
A	Adding a radioactive probe	To bind to all DNA fragments for visualisation of the DNA bands
B	Adding restriction enzymes	To digest each DNA sample into fragments of similar lengths
C	Electrophoresis	To separate proteins based on the amount of negative charges the protein has
D	Transferring DNA fragments from a gel to a nitrocellulose paper with the use of alkaline solution	To permanently attach the DNA fragments to a surface and to separate the two complementary DNA strands

- 19** Rice crops in Japan are damaged by the green rice leafhopper (*Nephotettix cincticeps*), a pest that reduces crop yield.

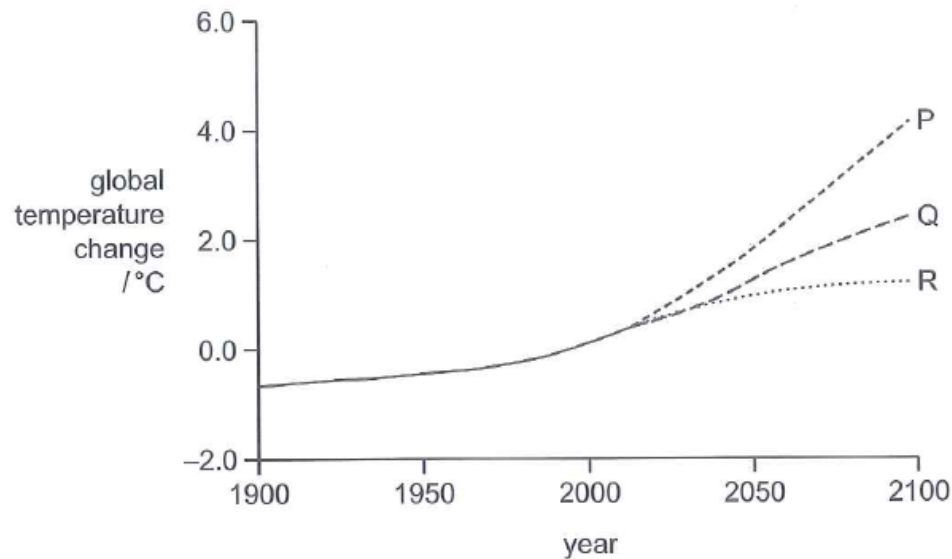
In a study of the effect of climate change on crop damage by the green rice leafhopper, it was found that an increase in winter temperatures caused an increase in crop damage, while an increase in summer temperatures caused a decrease in crop damage.

Which of the following are possible explanations for these findings?

- 1 Increased temperatures in the summer disrupt metabolic reactions in pests.
- 2 Increased temperatures in the summer cause a rise in the pests' metabolic rate that results shorter life cycle.
- 3 Increased temperatures in the winter disrupt the pests' life cycle and result in fewer being able to reproduce.
- 4 Increased temperatures in the winter allow more pests to survive and results in an increase in the pest population.

- A** 1 and 3 only
B 1 and 4 only
C 2 and 3 only
D 2 and 4 only

- 20 The graph shows the predicted change in global temperatures using three different models, P, Q and R. Model Q assumes that no new factors act to influence the rate of climate change.



The predictions of models P and R can be explained using some of the following statements.

- 1 Expansion of rainforest.
- 2 Increase in frequency of forest fires.
- 3 Permanently frozen soil and sediment in the Arctic begin to thaw.
- 4 Rising sea temperatures will reduce the solubility of greenhouse gases in the oceans.

Which of these statements support prediction of models P and R?

	Statements that support prediction P	Statements that support prediction R
A	2, 3 and 4	1
B	1 and 3	2 and 4
C	2	1, 3 and 4
D	3 and 4	1 and 2

21 Tamiflu is an antiviral drug used to treat influenza A (H1N1) patients. It works by preventing progeny virions from detaching from infected cells. Which of the following components of the influenza virus does Tamiflu act on?

- A** Haemagglutinin
- B** Matrix protein
- C** Neuraminidase
- D** Nucleoproteins

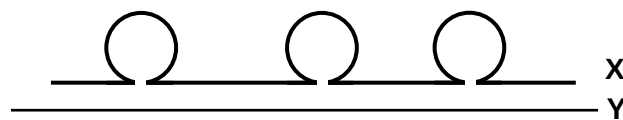
22 Which statement correctly describe the transfer of DNA in bacteria?

	Binary fission	Transduction	Conjugation
A	Bacterial chromosome and plasmids pass to daughter cells.	DNA transferred by bacteriophage from one bacterium to another.	Single strand of F plasmid transferred from one bacterium to another.
B	Bacterial chromosome and plasmids pass to daughter cells.	DNA transferred by bacteriophage from one bacterium to another.	Double strand of F plasmid transferred from one bacterium to another.
C	Only plasmids pass to daughter cells.	Bacterial cells take up foreign DNA from culture medium.	Single strand of F plasmid transferred from one bacterium to another.
D	Only plasmids pass to daughter cells.	Bacterial cells take up foreign DNA from culture medium.	Double strand of F plasmid transferred from one bacterium to another.

23 Which best describes the correct order (from earliest to latest) in the control of gene expression in protein synthesis?

- A** Polyribosome, promoter, spliceosome, histone, primary transcript
- B** Spliceosome, histone, promoter, primary transcript, polyribosome
- C** Histone, promoter, primary transcript, spliceosome, polyribosome
- D** Primary transcript, histone, spliceosome, promoter, polyribosome

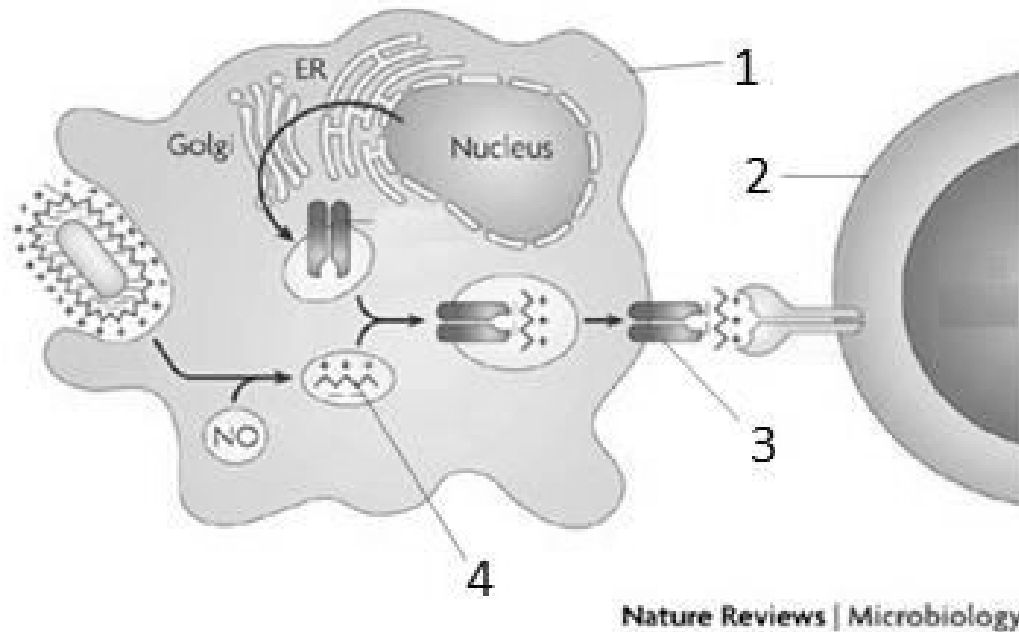
24 The diagram below shows the result of hybridizing mature mRNA of ovalbumin to the ovalbumin gene.



Which of the following statement is correct?

	X	Y	Number of exons	Number of introns
A	Mature mRNA	DNA	3	4
B	Mature mRNA	DNA	4	3
C	DNA	Mature mRNA	3	4
D	DNA	Mature mRNA	4	3

25 The figure below shows a process in an adaptive immune response.



Nitric oxide (NO) is a reactive substance that destroys microbes.

Which one of the followings best shows the labelling of structures 1 to 4?

	1	2	3	4
A	Virus infected Dendritic cell	CD8 T cells	Class I MHC	Antigen
B	Virus infected Dendritic cell	CD8 T cells	Antigen C	Class II MH
C	Dendritic cell	CD4 T cells	Antigen	Class I MHC
D	Dendritic cell	CD4 T cells	Class II MHC	Antigen

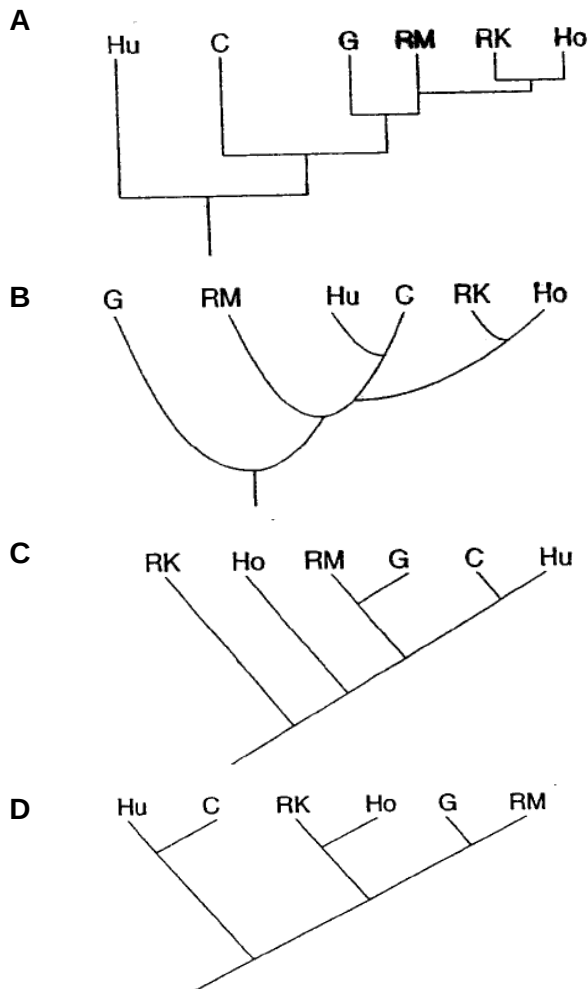
26 The DNA for a heavy chain in a B-cell making IgG₄ antibody specific for retroviruses has the following structure: 5'-V₁₄D₃J₂C_{γ4}-C_ε-C_{α1}-C_{α2}-3'. How many individual rearrangements were required to go from the embryonic DNA to this B-cell DNA?

- A 1
- B 2
- C 3
- D 4

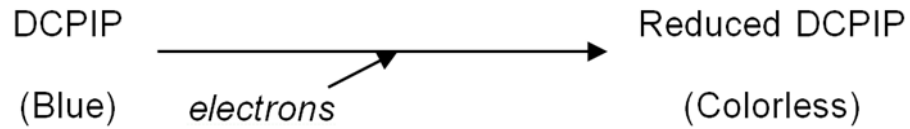
- 27 The table below shows the amino acid sequence of part of the beta-haemoglobin molecule found in 6 different vertebrates. The beta-globin chain has a total of 146 amino acids. Only the amino acids found in positions 101 to 114 are represented.

Amino acid position	101	102	103	104	105	106	107	108	109	110	111	112	113	114
Human (Hu)	Asn	Phe	Arg	Leu	Gly	Asn	Val	Leu	Val	Cys	Val	Leu	Ala	His
Red Kangaroo (RK)	Asn	Phe	Lys	Leu	Gly	Asn	Ile	Ile	Val	Ile	Cys	Leu	Ala	Glu
Rhesus Monkey (RM)	Asn	Phe	Lys	Leu	Gly	Asn	Val	Leu	Val	Cys	Val	Leu	Ala	His
Gorilla (G)	Asn	Phe	Lys	Leu	Gly	Asn	Val	Leu	Val	Cys	Val	Leu	Ala	His
Chimpanzee (C)	Asn	Phe	Arg	Leu	Gly	Asn	Val	Leu	Val	Cys	Val	Leu	Ala	His
Horse (Ho)	Asn	Phe	Arg	Leu	Gly	Asn	Val	Leu	Ala	Leu	Val	Val	Ala	Arg

Which of the following phylogenetic diagram best represents the evolutionary relationship of the vertebrates above?



- 28** 2,6-dichlorophenolindophenol (DCPIP) is a blue dye that can be converted to colourless reduced DCPIP by accepting electrons, as shown as follows.

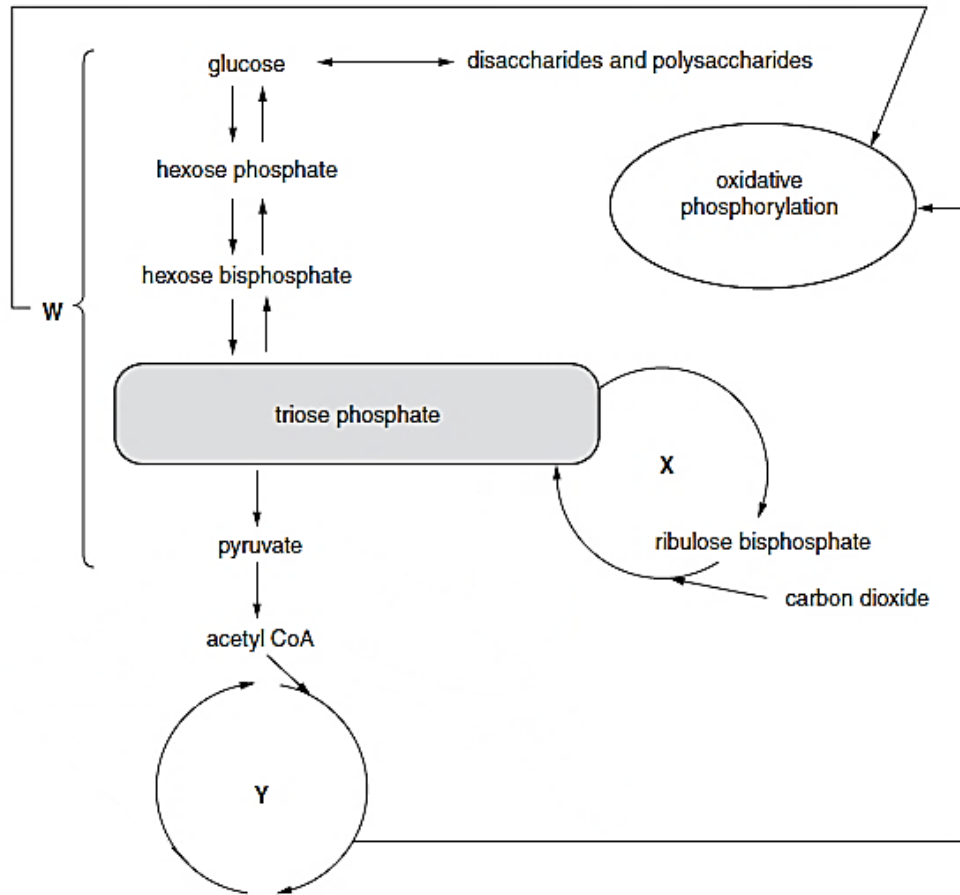


In an experiment, green chloroplast extract was first mixed with DCPIP and the extract turned blue-green. After exposure to 2 hours of light in the presence of fixed volumes of carbon dioxide and water, the extract became completely green again.

Which of the following shows a likely combination of products that was being formed towards the end of the experiment?

	O₂	ATP	reduced NADP
A	+	+	–
B	+	+	+
C	–	–	–
D	–	–	+

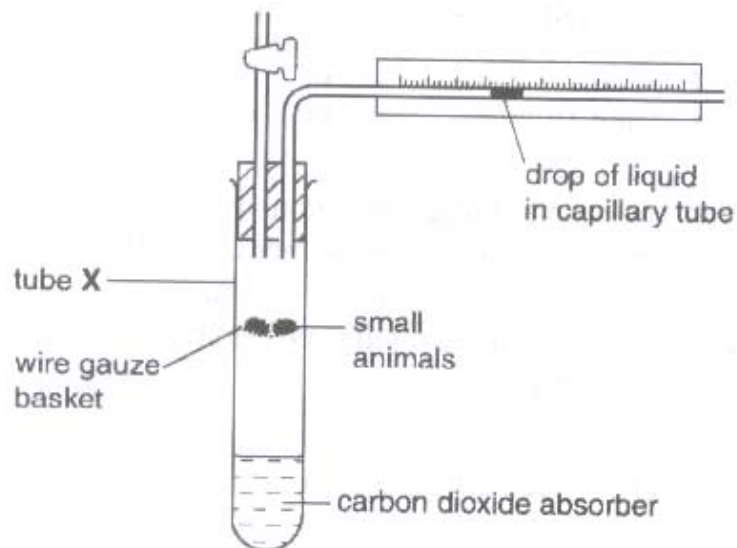
29 The diagram represents some of the reactions that take place in a leaf cell.



Which statement explains why the three reaction pathways, W, X and Y, are able to work concurrently in the same leaf cell?

- A Only X and Y can take place in the absence of oxygen.
- B W, X and Y are separated by membranes, allowing for the formation of separate proton gradients to synthesise ATP.
- C W, X and Y are separated by membranes, allowing for the maintenance of different conditions for enzymes to function.
- D X occurs independently of light unlike W and Y.

30 The diagram shows a simple respirometer.



The changes in gas volume in the tube are measured at intervals.

Time (minutes)	Gas volume with carbon dioxide absorber (cm ³)	Gas volume without carbon dioxide absorber (cm ³)
0	0.0	0.0
10	-0.4	-0.1
20	-0.8	-0.2
30	-1.2	-0.3

Tube X contains 2 g of small animals.

What is the carbon dioxide output per g per hour for these organisms?

- A 0.9 cm³
- B 1.8 cm³
- C 2.4 cm³
- D 4.8 cm³

END OF PAPER

2018 Y6 Preliminary Exam H2

MCQ Answer Scheme

1	D	16	A
2	A	17	A
3	C	18	D
4	C	19	B
5	A	20	A
6	C	21	C
7	A	22	A
8	A	23	C
9	D	24	D
10	D	25	D
11	D	26	C
12	C	27	C
13	B	28	A
14	C	29	C
15	A	30	A